

CHAPTER 6: HARMFUL MICROORGANISMS IN MILK AND DAIRY PRODUCTS I

6.1. Escherichia coli

Escherichia coli is a Gram-negative, rod-shaped, facultatively anaerobic bacillus belonging to the Enterobacteriaceae family. In dairy microbiology, *E. coli* serves as the universal index of fecal contamination and hygienic failure.

Because *E. coli* is normal flora in the gastrointestinal tract of warm-blooded animals, its detection in pasteurized dairy products indicates bad milking sanitation, dirty water, or poor post-pasteurization packaging hygiene.

Pathogenic strains represent serious public health threats:

- **Enterotoxigenic *E. coli* (ETEC):** Produces enterotoxins that cause watery diarrhea, particularly in children.
- **Enterohemorrhagic *E. coli* (EHEC, e.g., O157:H7):** A major pathogen that can contaminate raw milk from manure-covered cattle hides. It produces Shiga-like toxins that bind to intestinal receptors, causing severe bloody diarrhea, colitis, and can lead to Hemolytic Uremic Syndrome (HUS), characterised by acute renal failure and hemolysis. Consequently, complete zero-tolerance regulatory limits are enforced for *E. coli* in commercial pasteurized milk.

6.2. Salmonella

Salmonella is a genus of rod-shaped, motile, Gram-negative, non-sporeforming Enterobacteriaceae. The primary species of interest in milk safety are *Salmonella enterica* serovars Typhimurium, Enteritidis, and Dublin.